Geometric Quantities for the FLRW Metric

Generated by Einsteinpy_to_latex

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- Einstein, A. (1915). Die Feldgleichungen der Gravitation. *Sitzungsberichte der Königlich Preußischen Akademie der Wissenschaften*, 844-847.
- Ribeiro, S., Bapat, A., Tandon, A., & EinsteinPy Developers (2020). EinsteinPy: A Python Package for General Relativity. *Journal of Open Source Software*, 5(51), 2356. `https://doi.org/10.21105/joss.02356`
- P. Ntelis (2026). Einsteinpy_to_latex: FLRW Metric Tensor Calculator (Version 1.0). `https://github.com/lontelis/Einsteinpy_to_latex`

1 Geometric quantities for the FLRW metric

The metric is given by

$$ds^2 = -c^2dt^2 + a(t)^2(dx^2 + dy^2 + dz^2).$$

1.1 Metric tensor $g_{\mu\nu}$

$$g_{\mu\nu} = \begin{bmatrix} -c^2 & 0 & 0 & 0 \\ 0 & a^2(t) & 0 & 0 \\ 0 & 0 & a^2(t) & 0 \\ 0 & 0 & 0 & a^2(t) \end{bmatrix} \tag{1}$$

1.2 Christoffel symbols $\Gamma^{\mu}_{\nu\rho}$ (non-zero)

$$\Gamma^{t}_{xx} = \frac{a(t)\frac{d}{dt}a(t)}{c^2} \tag{2}$$

$$\Gamma^{t}_{yy} = \frac{a(t)\frac{d}{dt}a(t)}{c^2} \tag{3}$$

$$\Gamma^{t}_{zz} = \frac{a(t)\frac{d}{dt}a(t)}{c^2} \tag{4}$$

$$\Gamma^{x}_{tx} = \frac{\frac{d}{dt}a(t)}{a(t)} \tag{5}$$

$$\Gamma^{x}_{xt} = \frac{\frac{d}{dt}a(t)}{a(t)} \tag{6}$$

$$\Gamma^{y}_{ty} = \frac{\frac{d}{dt}a(t)}{a(t)} \tag{7}$$

$$\Gamma^{y}_{yt} = \frac{\frac{d}{dt}a(t)}{a(t)} \tag{8}$$

$$\Gamma^{z}_{tz} = \frac{\frac{d}{dt}a(t)}{a(t)} \tag{9}$$

$$\Gamma^{z}_{zt} = \frac{\frac{d}{dt}a(t)}{a(t)} \tag{10}$$

$$\tag{11}$$

1.3 Riemann curvature tensor $R^{\mu}{}_{\nu\rho\sigma}$ (non-zero)

$$R^{t}_{xtx} = \frac{a(t)\frac{d^2}{dt^2}a(t)}{c^2} \tag{12}$$

$$R^{t}_{xxt} = -\frac{a(t)\frac{d^2}{dt^2}a(t)}{c^2} \tag{13}$$

$$R^{t}_{yty} = \frac{a(t)\frac{d^2}{dt^2}a(t)}{c^2} \tag{14}$$

$$R^{t}_{yyt} = -\frac{a(t)\frac{d^2}{dt^2}a(t)}{c^2} \tag{15}$$

$$R^{t}_{ztz} = \frac{a(t)\frac{d^2}{dt^2}a(t)}{c^2} \tag{16}$$

$$R^{t}_{zzt} = -\frac{a(t)\frac{d^2}{dt^2}a(t)}{c^2} \tag{17}$$

$$R^{x}_{ttx} = \frac{\frac{d^2}{dt^2}a(t)}{a(t)} \tag{18}$$

$$R^{x}_{txt} = -\frac{\frac{d^2}{dt^2}a(t)}{a(t)} \tag{19}$$

$$R^{x}_{yxy} = \frac{\left(\frac{d}{dt}a(t)\right)^2}{c^2} \tag{20}$$

$$R^{x}_{yyx} = -\frac{\left(\frac{d}{dt}a(t)\right)^2}{c^2} \tag{21}$$

$$R^{x}_{zxz} = \frac{\left(\frac{d}{dt}a(t)\right)^2}{c^2} \tag{22}$$

$$R^{x}_{zzx} = -\frac{\left(\frac{d}{dt}a(t)\right)^2}{c^2} \tag{23}$$

$$R^{y}_{tty} = \frac{\frac{d^2}{dt^2}a(t)}{a(t)} \tag{24}$$

$$R^{y}_{tyt} = -\frac{\frac{d^2}{dt^2}a(t)}{a(t)} \tag{25}$$

$$R^{y}_{xxy} = -\frac{\left(\frac{d}{dt}a(t)\right)^2}{c^2} \tag{26}$$

$$R^{y}_{xyx} = \frac{\left(\frac{d}{dt}a(t)\right)^2}{c^2} \tag{27}$$

$$R^{y}_{zyz} = \frac{\left(\frac{d}{dt}a(t)\right)^2}{c^2} \tag{28}$$

$$R^{y}_{zzy} = -\frac{\left(\frac{d}{dt}a(t)\right)^2}{c^2} \tag{29}$$

$$R^{z}_{ttz} = \frac{\frac{d^2}{dt^2}a(t)}{a(t)} \tag{30}$$

$$R^{z}_{tzt} = -\frac{\frac{d^2}{dt^2}a(t)}{a(t)} \tag{31}$$

$$R^{z}_{xxz} = -\frac{\left(\frac{d}{dt}a(t)\right)^2}{c^2} \tag{32}$$

$$R^{z}_{xzx} = \frac{\left(\frac{d}{dt}a(t)\right)^2}{c^2} \tag{33}$$

$$R^{z}_{yyz} = -\frac{\left(\frac{d}{dt}a(t)\right)^2}{c^2} \tag{34}$$

$$R^{z} = \left(\frac{d}{dt}a(t)\right)^2 \tag{35}$$

1.4 Ricci tensor $R_{\mu\nu}$ (non-zero)

$$R_{tt} = -\frac{3\frac{d^2}{dt^2}a(t)}{a(t)} \tag{37}$$

$$R_{xx} = \frac{a(t)\frac{d^2}{dt^2}a(t) + 2\left(\frac{d}{dt}a(t)\right)^2}{c^2} \tag{38}$$

$$R_{yy} = \frac{a(t)\frac{d^2}{dt^2}a(t) + 2\left(\frac{d}{dt}a(t)\right)^2}{c^2} \tag{39}$$

$$R_{zz} = \frac{a(t)\frac{d^2}{dt^2}a(t) + 2\left(\frac{d}{dt}a(t)\right)^2}{c^2} \tag{40}$$

$$\tag{41}$$

1.5 Ricci scalar R

$$\frac{6\left(a(t)\frac{d^2}{dt^2}a(t) + \left(\frac{d}{dt}a(t)\right)^2\right)}{c^2a^2(t)} \tag{42}$$

1.6 Einstein tensor $G_{\mu\nu}$ (non-zero)

$$G_{tt} = \frac{3.0\left(\frac{d}{dt}a(t)\right)^2}{a^2(t)} \tag{43}$$

$$G_{xx} = \frac{-2.0a(t)\frac{d^2}{dt^2}a(t) - 1.0\left(\frac{d}{dt}a(t)\right)^2}{c^2} \tag{44}$$

$$G_{yy} = \frac{-2.0a(t)\frac{d^2}{dt^2}a(t) - 1.0\left(\frac{d}{dt}a(t)\right)^2}{c^2} \tag{45}$$

$$G_{zz} = \frac{-2.0a(t)\frac{d^2}{dt^2}a(t) - 1.0\left(\frac{d}{dt}a(t)\right)^2}{c^2} \tag{46}$$

$$\tag{47}$$